1 Linear Equations in Linear Algebra

1.3

VECTOR EQUATIONS

PEARSON

Vectors in $\mathbb{R}^2$

- A matrix with only one column is called a **column vector**, or simply a **vector**.
- An example of a vector with two entries is

$$\mathrm{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix},$$

where w_1 and w_2 are any real numbers.

- The set of all vectors with two entries is denoted by $\mathbb{R}^2$ (read "r-two").

- The $\mathbb{R}$ stands for the real numbers that appear as entries in the vector, and the exponent 2 indicates that each vector contains two entries.
- Two vectors in $\mathbb{R}^2$ are **equal** if and only if their corresponding entries are equal.
- Given two vectors **u** and **v** in $\mathbb{R}^2$, their **sum** is the vector $\mathrm{u}+\mathrm{v}$ obtained by adding corresponding entries of **u** and **v**.
- Given a vector **u** and a real number c, the **scalar multiple** of **u** by c is the vector $c\mathbf{u}$ obtained by multiplying each entry in **u** by c.

Example 1: Given $\mathrm{u}=\begin{bmatrix}1\\-2\end{bmatrix}$ and $\mathrm{v}=\begin{bmatrix}2\\-5\end{bmatrix}$, find

$4\mathbf{u}, (-3)\mathrm{v}$, and $4\mathrm{u}+(-3)\mathrm{v}$.

Solution: $4\mathrm{u}=\begin{bmatrix}4\\-8\end{bmatrix}$, $(-3)\mathrm{v}=\begin{bmatrix}-6\\15\end{bmatrix}$ and

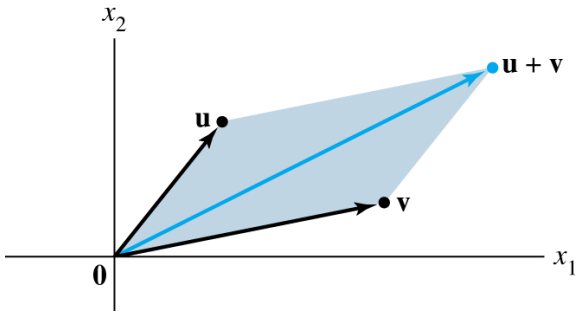
$$4\mathrm{u}+(-3)\mathrm{v}=\begin{bmatrix}4\\-8\end{bmatrix}+\begin{bmatrix}-6\\15\end{bmatrix}=\begin{bmatrix}-2\\7\end{bmatrix}$$

GEOMETRIC DESCRIPTIONS OF $\mathbb{R}^2$

- Consider a rectangular coordinate system in the plane. Because each point in the plane is determined by an ordered pair of numbers, *we can identify a geometric point (a, b) with the column vector* $\begin{bmatrix} a \\ b \end{bmatrix}$.
- So we may regard $\mathbb{R}^2$ as the set of all points in the plane.

PARALLELOGRAM RULE FOR ADDITION

- If **u** and **v** in $\mathbb{R}^2$ are represented as points in the plane, then $\mathbf{u} + \mathbf{v}$ corresponds to the fourth vertex of the parallelogram whose other vertices are **u**, **0**, and **v**. See Fig. 3 below.



VECTORS IN $\mathbb{R}^3$ and $\mathbb{R}^n$

- Vectors in $\mathbb{R}^3$ are 3×1 column matrices with three entries.
- They are represented geometrically by points in a three-dimensional coordinate space, with arrows from the origin sometimes included for visual clarity.
- If n is a positive integer, $\mathbb{R}^n$ (read "r-n") denotes the collection of all lists (or *ordered n-tuples*) of n real numbers, usually written as $n\times1$ column matrices, such as

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}.$$

ALGEBRAIC PROPERTIES OF $\mathbb{R}^n$

- The vector whose entries are all zero is called the **zero vector** and is denoted by **0**.
- For all **u**, **v**, **w** in $\mathbb{R}^n$ and all scalars c and d:

(i) $\mathbf{u}+\mathbf{v}=\mathbf{v}+\mathbf{u}$

(ii) $(\mathbf{u}+\mathbf{v})+\mathbf{w}=\mathbf{u}+(\mathbf{v}+\mathbf{w})$

(iii) $\mathbf{u}+0=0+\mathbf{u}=\mathbf{u}$

(iv) $\mathbf{u}+(-\mathbf{u})=-\mathbf{u}+\mathbf{u}=0$, where $-\mathbf{u}$ denotes $(-1)\mathbf{u}$

(v) $c(\mathbf{u}+\mathbf{v})=c\mathbf{u}+c\mathbf{v}$

(vi) $(c+d)\mathbf{u}=c\mathbf{u}+d\mathbf{u}$

- (vii) $c(d\mathrm{u})=(\mathrm{cd})(\mathrm{u})$
  (viii) $1\mathrm{u}=\mathrm{u}$

- Given vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p$ in $\mathbb{R}^n$ and given scalars $c_1, c_2, \ldots, c_p$, the vector $\mathbf{y}$ defined by

$$\mathrm{y}=c_1\mathrm{v}_1+\ldots+c_p\mathrm{v}_p$$

  is called a **linear combination** of $\mathbf{v}_1, \ldots, \mathbf{v}_p$ with **weights** $c_1, \ldots, c_p$.

- The weights in a linear combination can be any real numbers, including zero.

- **Example 2:** Let $\mathrm{a}_1=\begin{bmatrix}1\\-2\\-5\end{bmatrix}$, $\mathrm{a}_2=\begin{bmatrix}2\\5\\6\end{bmatrix}$ and $\mathrm{b}=\begin{bmatrix}7\\4\\-3\end{bmatrix}$.

  Determine whether **b** can be generated (or written) as a linear combination of $\mathbf{a}_1$ and $\mathbf{a}_2$. That is, determine whether weights x_1 and x_2 exist such that

$$x_1\mathrm{a}_1+x_2\mathrm{a}_2=\mathrm{b} \qquad (1)$$

  If vector equation (1) has a solution, find it.

Solution: Use the definitions of scalar multiplication and vector addition to rewrite the vector equation

$$x_1\begin{bmatrix}1\\-2\\-5\end{bmatrix}+x_2\begin{bmatrix}2\\5\\6\end{bmatrix}=\begin{bmatrix}7\\4\\-3\end{bmatrix},$$

$\mathbf{a}_1$ $\mathbf{a}_2$ $\mathbf{b}$

which is same as $\begin{bmatrix}x_1\\-2x_1\\-5x_1\end{bmatrix}+\begin{bmatrix}2x_2\\5x_2\\6x_2\end{bmatrix}=\begin{bmatrix}7\\4\\-3\end{bmatrix}$

and $\begin{bmatrix}x_1+2x_2\\-2x_1+5x_2\\-5x_1+6x_2\end{bmatrix}=\begin{bmatrix}7\\4\\-3\end{bmatrix}.$ (2)

- The vectors on the left and right sides of (2) are equal if and only if their corresponding entries are both equal. That is, x_1 and x_2 make the vector equation (1) true if and only if x_1 and x_2 satisfy the following system.

$$\begin{aligned}x_1+2x_2&=7\\-2x_1+5x_2&=4\\-5x_1+6x_2&=-3\end{aligned}\qquad(3)$$

- To solve this system, row reduce the augmented matrix of the system as follows:

$$\begin{bmatrix} 1 & 2 & 7 \\ -2 & 5 & 4 \\ -5 & 6 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 7 \\ 0 & 9 & 18 \\ 0 & 16 & 32 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 7 \\ 0 & 1 & 2 \\ 0 & 16 & 32 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

- The solution of (3) is $x_1 = 3$ and $x_2 = 2$. Hence $\mathbf{b}$ is a linear combination of $\mathbf{a}_1$ and $\mathbf{a}_2$, with weights $x_1 = 3$ and $x_2 = 2$. That is,

$$3\begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix} + 2\begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}.$$

- Now, observe that the original vectors $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{b}$ are the columns of the augmented matrix that we row reduced:

$$\begin{bmatrix} 1 & 2 & 7 \\ -2 & 5 & 4 \\ -5 & 6 & -3 \end{bmatrix}$$

$\mathbf{a}_1$ $\mathbf{a}_2$ $\mathbf{b}$

- Write this matrix in a way that identifies its columns.

$$\begin{bmatrix} a_1 & a_2 & b \end{bmatrix} \qquad (4)$$

- A vector equation

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \ldots + x_n\mathbf{a}_n = \mathbf{b}$$

has the same solution set as the linear system whose augmented matrix is

$$\begin{bmatrix}\mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n & \mathbf{b}\end{bmatrix} \qquad (5)$$

- In particular, **b** can be generated by a linear combination of $\mathbf{a}_1, \ldots, \mathbf{a}_n$ if and only if there exists a solution to the linear system corresponding to the matrix (5).

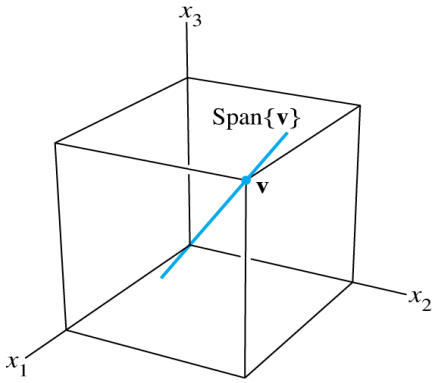
- **Definition:** If $\mathbf{v}_1, \ldots, \mathbf{v}_p$ are in $\mathbb{R}^n$, then the set of all linear combinations of $\mathbf{v}_1, \ldots, \mathbf{v}_p$ is denoted by Span $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ and is called the **subset of $\mathbb{R}^n$ spanned** (or **generated**) **by** $\mathbf{v}_1, \ldots, \mathbf{v}_p$. That is, Span $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ is the collection of all vectors that can be written in the form

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots + c_p\mathbf{v}_p$$

with $c_1, \ldots, c_p$ scalars.

A GEOMETRIC DESCRIPTION OF SPAN {V}

- Let **v** be a nonzero vector in $\mathbb{R}^3$. Then Span {**v**} is the set of all scalar multiples of **v**, which is the set of points on the line in $\mathbb{R}^3$ through **v** and **0**. See Fig. 10 below:



A GEOMETRIC DESCRIPTION OF SPAN {U, V}

- If **u** and **v** are nonzero vectors in $\mathbb{R}^3$, with **v** not a multiple of **u**, then Span {**u, v**} is the plane in $\mathbb{R}^3$ that contains **u**, **v**, and **0**.
- In particular, Span {**u**, **v**} contains the line in $\mathbb{R}^3$ through **u** and **0** and the line through **v** and **0**. See Fig. 11 below:

